

Light – Reflection and Refraction (Set A) — MCQ Worksheet

Science • Chapter 9 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The angle of incidence is equal to the:

- (A) Angle of reflection
- (B) Angle of refraction
- (C) Critical angle
- (D) Angle of deviation

Q2. Focal length of a spherical mirror is related to its radius of curvature R as:

- (A) $f = 2R$
- (B) $f = R$
- (C) $f = R/2$
- (D) $f = R/4$

Q3. A concave mirror always gives a virtual, erect image when the object is placed:

- (A) At infinity
- (B) At C
- (C) Beyond C
- (D) Between F and pole

Q4. A convex mirror always forms an image that is:

- (A) Real and inverted
- (B) Virtual, erect, diminished
- (C) Real and magnified
- (D) Virtual and magnified

Q5. Convex mirrors are used as rear-view mirrors because they:

- (A) Magnify the image
- (B) Give a wide field of view
- (C) Form real images
- (D) Produce no parallax

Q6. Mirror formula is:

- (A) $1/v - 1/u = 1/f$
- (B) $1/v + 1/u = 1/f$
- (C) $v + u = f$
- (D) $v - u = f$

Q7. Magnification produced by a mirror is given by:

- (A) $m = u/v$
- (B) $m = -v/u$
- (C) $m = v/u$
- (D) $m = u + v$

Q8. In sign convention, distances measured against the direction of incident light are taken as:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinite

Q9. When light goes from a rarer to a denser medium, it bends:

- (A) Towards the normal
- (B) Away from the normal
- (C) Parallel to the normal
- (D) Does not bend

Q10. Snell's law: $n = \sin i / \sin r$ is also called the law of:

- (A) Reflection
- (B) Refraction
- (C) Dispersion
- (D) Total internal reflection

Q11. Refractive index of a medium $n = c/v$ depends on the:

- (A) Angle of incidence
- (B) Speed of light in the medium
- (C) Mass of the medium
- (D) Density only

Q12. A glass slab causes:

- (A) No bending of light
- (B) Lateral displacement only
- (C) Total internal reflection
- (D) Reflection only

Q13. A convex lens converges parallel rays to a:

- (A) Virtual focus
- (B) Real focus
- (C) Both
- (D) Neither

Q14. A concave lens always gives an image that is:

- (A) Real and inverted
- (B) Virtual, erect, diminished
- (C) Magnified
- (D) Same size

Q15. The lens formula is:

(A) $1/v + 1/u = 1/f$

(C) $v + u = f$

(B) $1/v - 1/u = 1/f$

(D) $v \times u = f$

Q16. The unit of power of a lens is:

(A) Metre

(C) Dioptre

(B) Watt

(D) Joule

Q17. The power of a convex lens is:

(A) Negative

(C) Zero

(B) Positive

(D) Infinite

Q18. Power of a lens whose focal length is 50 cm is:

(A) +0.5 D

(C) +5 D

(B) +2 D

(D) +50 D

Q19. A magnifying glass uses a:

(A) Concave lens

(C) Plane mirror

(B) Convex lens with object within F

(D) Concave mirror

Q20. When the object is placed at the centre of curvature of a concave mirror, the image is:

(A) Real, inverted, same size

(C) Real, inverted, diminished

(B) Virtual, erect, magnified

(D) Virtual, inverted, magnified

Q21. Magnification produced by a lens is:

(A) $m = -v/u$

(C) $m = u/v$

(B) $m = v/u$

(D) $m = u-v$

Q22. For a concave lens, the focal length is taken as:

(A) Positive

(C) Zero

(B) Negative

(D) Infinite

Q23. When the object is at infinity for a concave mirror, the image is formed:

(A) At F, real, inverted, highly diminished

(C) At pole

(B) At C, real, magnified

(D) Between F and pole

Q24. The two laws of reflection are valid for:

(A) Plane mirrors only

(C) Convex mirrors only

(B) Concave mirrors only

(D) All types of mirrors

Q25. A real image:

(A) Cannot be formed on a screen

(C) Is always erect

(B) Can be formed on a screen

(D) Is always magnified